RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management and Economics		**Course Code : HS261TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List the five important functions of management.	2	1	1
2.	Identify which statement represents vision and which represents mission: i. "To be a globally recognized brand known for innovation." ii. "We design and deliver innovative products to meet customer needs."	2	1	1
3.	Do you agree that coordination need not be treated as a separate function of management? Give a brief reason.	2	2	1
4.	Explain how economists respond to the criticism that people often make irrational decisions due to emotions, habits, or lack of information.	2	2	3
5.	Explain how the government is involved in the circular flow of income.	2	2	3
	Part – B			
1	A manager at Swiggy handles team coordination, information sharing, and decision-making during peak hours. Apply Henry Mintzberg's managerial roles to illustrate how the manager performs these functions in this situation.	10	L3	1
2	In the launch of a new electric vehicle by Tata Motors, the project manager must coordinate teams, solve problems, and motivate employees. Apply the important managerial skills to show how the manager can successfully handle this project.	10	L3	1
3	Using a real-life example, explain the concept of circular flow of income to show the role of households, firms, government, and foreign sector.	10	L2	3
4	Explain the basic economic problem of scarcity and choice and describe how individuals and firms allocate limited resources.	10	L2	3
5	Describe the different types of economic systems and explain any two with examples.	10	L2	3

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Date : 13/04/2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Supply Chain Management		**Course Code : IM362IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part–A			
1.	State the objective of a supply chain.	1	1	CO1
2.	List the typical stages in a supply chain.	2	1	CO1
3.	_____ cycle occurs between the manufacturer and the supplier.	1	1	CO1
4.	Write the full form of the following supply chain macro processes: CRM, ISCM, SRM	2	2	CO1
5.	What is a company's supply chain strategy?	2	1	CO1
6.	Illustrate the relationship between number of facilities and total logistics cost.	2	2	CO2
	Part–B			
1	Explain the impact of supply chain decision phases on the overall profitability and success of manufacturing supply chains.	10	3	CO1
2	Illustrate and discuss cycle view of an automobile component manufacturing organization.	10	3	CO1
3	List the three basic steps to achieving this strategic fit. Evaluate the first step.	10	2	CO1
4	Unipart, a manufacturer of auto parts, is considering two B2B marketplaces to purchase its MRO supplies. Both marketplaces offer a full line of supplies at very similar prices for products and shipping. Both provide similar service levels and lead times. However, their fee structures are quite different. The first marketplace, Parts4u.com, sells all of its products with a 5 percent commission tacked on top of the price of the product (not including shipping). AllMRO.com's pricing is based on a subscription fee of $10 million that must be paid up front for a two-year period and a commission of 1 percent on each transaction's product price Unipart spends about $150 million on MRO supplies each year, although this varies with their utilization. Next year will likely be a strong year, in which high utilization will keep MRO spending at $150 million. However, there is a 25 percent chance that spending will drop by 10 percent. The second year, there is a 50 percent chance that the spending level will stay where it was in the first year and a 50 percent chance that it will drop by another 10 percent. Unipart uses a discount rate of 20 percent. Assume all costs are incurred at the beginning of each year (so Year 1 costs are incurred now and Year 2 costs are incurred in a year). Which B2B marketplace should Unipart buy its parts from?	20	3	CO2

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Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : 6th	**UG**	**Duration : 30 + 90 Min**
Course Title: ERGONOMICS		**Course Code : IM363IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define the term ergonomics.	01	L_1	CO1
2.	Name any two benefits of ergonomics	01	L_2	CO1
3.	What is meant by human centred design?	01	L_1	CO1
4.	Mention one example of cognitive ergonomics while interfacing with websites?.	01	L_2	CO1
5.	List Give a practical example of a Functional Anthropometric Data.	01	L_2	CO2
6.	What are the two important landmark percentiles in the normal distribution are as applied to consideration of anthropometric variables in designs?	01	L_2	CO2
7.	State any one principle of workspace design for sitting tasks.	01	L_2	CO2
8.	What is the primary goal of anthropometry?	01	L_2	CO2
9.	Give an example of application of Newtonian anthropometrics in designs	01	L_3	CO2
10.	Expand the acronyms CTD.	01	L_1	CO2
	Part – B			
1.	Develop a step-by-step ergonomic evaluation plan for a computer based VDT based workstation. Your answer should include: Identification of key task elements, Application of anthropometric measurements, Postural analysis methods, Recommendations to reduce musculoskeletal risks.	10	L_4	CO1
2.	Distinguish between Macro ergonomics and Micro ergonomics. Outline the scope of work that is covered in both these areas of ergonomics. Give an example to justify the application of these specific to work systems design and equipment/workspace designs. List any two factors to be considered for each of these two types of ergonomics in designs.	10	L_2	CO1
3.	Consider the example of a handle Design of a Handheld Power Tool: How should you calculate the Grip diameter based on the 5th and 95th percentile values of a hand breadth. For easy - Thumb reach: where should the controls be placed considering natural thumb arc. Which force can be reduced by choosing - Material: such as soft rubberized textures? Which of the two grips whether inline grip or pistol grip must be so as to minimize wrist deviation. If you can reduce the handle mass to $\leq$ 1 kg and achieve the balance centre near grip, what is the implications for operator fatigue during the power tool operations? (Provide Sketches to highlight the engineering anthropometric design and to visualize the handles design of a Handheld power tool)	10	L_4	CO2

4.	A company is designing a sitting assembly station for both male and female operators in a watch assembly line. Apply anthropometric principles to determine optimal work surface height, Analyse how human variability influences design decisions, Discuss trade-offs between productivity, comfort, and injury prevention in this work context. (Sketch your ideas with clarity of critical design variables)	10	L_3	CO2
5a.	What is “dynamic sitting,” and why is it preferred over maintaining a “perfect” upright posture?	05	L_2	CO1
b.	The term Maximum Holding Time (MHT) is defined as The longest duration a specific physical posture or force can be maintained before failure or unacceptable discomfort. Give a practical example to emphasize as to why this variable is important for equipment/workspace design based on your own exposure and experience.	05	L_3	CO2

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Date : 16th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Human Resource Management & Analytics		**Course Code : IM364TA**

Note:
Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	HR manager advises a department head on hiring. What type of authority is this?	01	L1	CO1
2.	A company uses online interviews for hiring. Which trend is this in HRM?	01	L2	CO1
3.	Which HR group focuses on strategic decision-making?	01	L1	CO1
4.	What is talent management?	01	L1	CO1
5.	What are the key components of a job description?	01	L1	CO1
6.	What is HR forecasting?	01	L1	CO2
7.	List the three steps in Succession planning?	02	L2	CO2
8.	List any two internal sources of candidates.	01	L1	CO2
9.	Define Job analysis	01	L1	CO1
	Part – B			
1.	Explain the major trends shaping Human Resource Management in detail.	10	L2	CO1
2.	Imagine you are working in the HR department of a hospital called *CarePlus Medical Center*. The hospital wants to improve its hiring process for the role of Staff Nurse. Prepare: a) A Job Description for the role of Staff Nurse b) A Job Specification for the same role	10	L4	CO1
3.	A manufacturing company wants to analyze a job that involves repetitive manual tasks on the shop floor. Which job analysis methods would you recommend? Justify your answer by comparing alternative methods.	10	L3	CO1
4.	An e-commerce company is expanding and needs to hire for marketing, customer support, data analysis, and logistics roles. Identify different external sources for recruiting candidates and explain how each aligns with the company's needs.	10	L3	CO2
5.	Imagine you are the HR manager of a multinational company that is about to launch a recruitment campaign for a new product development team. The team will consist of software developers, project managers, and quality assurance specialists. a) Outline the steps of the recruitment process you would follow to attract and hire the best talent for this team. b) Explain how you would ensure that the process is efficient and cost-effective. Discuss any tools or technologies you might use to streamline this recruitment process.	10	L4	CO2

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Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: FACILITIES PLANNING AND DESIGN		**Course Code : IM365TDA**

Note:
1. Answer all the Questions.

PART – A

SI. No.	Questions	M	BT	CO
1	State any two factors influencing plant location.	02	L2	CO1
2	Identify the suitable layout for a hospital and justify your choice.	02	L2	CO1
3	Which layout is suitable for job-order production? Give reason.	02	L1	CO2
4	Analyze any two causes of inadequate facilities planning with examples.	02	L1	CO1
5	Propose a basic facilities planning strategy for a new manufacturing firm.	02	L2	CO2

PART – B

SI. No.	Questions	M	BT	CO
1.	Explain the concept of facilities planning and discuss its significance in modern industries.	10	L1	CO1
2.	Differentiate between facilities planning process and strategic planning process in terms of meaning, scope, objective, time horizon and focus.	10	L2	CO2
3.	Explain the factors influencing plant location with suitable examples.	10	L3	CO3
4.	Recommend a suitable plant layout for a manufacturing firm and justify your choice with merits and demerits.	10	L4	CO4
5.	Potential locations A, B and C have the cost structure as shown below for producing a product expected to sell for Rs. 4250 per unit. Site | Fixed Cost/year (Rs) | Variable Cost/unit (Rs) A | 85,00,000 | 775 B | 65,00,000 | 2225 C | 70,00,000 | 1460 Find the most economical location for an expected volume of 6250 units per year.	10	L2	CO2

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Date : 16th April 2026	**CIE - 1**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the primary goal of Service Operations Management?	2	1	1
2.	Name two major challenges faced by service operations managers in today's dynamic environment.	2	2	2
3.	Identify any 4 tools that have been used on you to collect information to enhance customer satisfaction.	2	3	1
4.	What type of service process is typically used in a fast-food restaurant, and why?	2	3	2
5.	How does service operations management differ from manufacturing operations management?	2	2	2
	Part – B			
1	What are the different types of service processes? Cite the example of any service organization and highlight the different processes involved in that organization.	10	2	2
2	Apply the concept of the "Poka-Yoke" (mistake-proofing) methodology to a service setting of your choice. Explain how you would design both warning and control types of poka-yokes to manage the operational risks associated with customer variability and employee variability.	10	3	2
3	Explain "Service concept" using the tabular column as applied to a "New age engineering education institute".	10	5	3
4	You are tasked with launching a new, subscription-based "tele-health" service that connects patients with doctors via video calls 24/7. This service is highly intangible and faces significant variability in demand. Create a set of success metrics that specifically measure the "moments of truth" in the virtual interaction, addressing the challenge of measuring quality for an intangible product. Hint: Go up to 3 Moments of Truth starting from Zeroth. Use tabular column and avoid writing in paragraphs.	10	4	2
5	You are a Service Operations Manager of an online retail store. What challenges would you may face as the manager and what measures would you take to overcome them?	10	3	3

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Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Design of Experiments		**Course Code : IM365TDD**

Note:
1. Answer all the Questions.
2. Use of statistical tables permitted

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List two applications of design of experiments.	1	1	CO1
2.	What are the two advantages of replication?	2	1	CO1
3.	_____ is an example of larger-the-better type quality characteristic.	1	1	CO2
4.	According to Genichi Taguchi "Quality is the _____ a product causes to the society"	2	2	CO2
5.	Identify response, controllable and noise factors for gas welding process.	2	2	CO1
6.	What are the steps in the engineering design problem strategy?	2	2	CO2
	Part – B			
1	Enumerate the guidelines for the statistical approach in designing and analyzing an experiment.	10	2	CO1
2	Trace the history of statistical design. Elucidate the important stages.	10	3	CO1
3	Write the expressions for quality loss function of various quality characteristics. Illustrate with sketches. List the situations in which these quality characteristics are applicable.	10	2	CO2
4	Identify the quality control activities during various product realization steps. Comment on the ability of these activities to reduce effect of noise factors.	05	3	CO2

<table>
<tr>
<td>5</td>
<td>An industrial engineer employed by a beverage bottler is interested in the effects of two different types of 32-ounce bottles (A) on the time to deliver 12-bottle cases of the product. The two bottle types are glass and plastic. Two workers (B) are used to perform a task consisting of moving 40 cases of the product 50 feet on a standard type of hand truck and stacking the cases in a display. Four replicates of a 2^2 factorial design are performed, and the times observed are listed in the following table:
<table>
<tr><td rowspan="2">Treatment Combination</td><td colspan="4">Replicates</td></tr>
<tr><td>I</td><td>II</td><td>III</td><td>IV</td></tr>
<tr><td>1</td><td>5.12</td><td>4.89</td><td>4.98</td><td>5.00</td></tr>
<tr><td>a</td><td>6.65</td><td>6.24</td><td>5.49</td><td>5.55</td></tr>
<tr><td>b</td><td>4.95</td><td>4.43</td><td>4.27</td><td>4.25</td></tr>
<tr><td>ab</td><td>5.28</td><td>4.91</td><td>4.75</td><td>4.71</td></tr>
</table>
i. Estimate the factor effects.
ii. Prepare an analysis of variance table and determine which factors are important in explaining time to deliver.</td>
<td>15</td>
<td>2</td>
<td>CO4</td>
</tr>
</table>

Academic year 2025-2026 (EVEN Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**16th April 2026**	**Maximum Marks**	**10 + 50**
Course Code	**IM266TEQ**	**Duration**	**120 Mins**
Sem	**VI Semester**	**CIE - I**	
ELEMENTS OF FINANCIAL MANAGEMENT (Institutional Elective)			

Note:
1. Answer all the Questions.
2. Use of Interest factor tables permitted.

PART – A

S No	Questions	M	BT	CO
1	Distinguish between Financial Assets and Liabilities.	2	2	2
2	Mention the role of SEBI in investor protection.	2	2	2
3	What is meant by the term, "Bottom Line" in the context of financial management?	2	2	1
4	What is meant by the term Depreciation? Mention how it is treated in preparation of Accounts.	2	2	1
5	Distinguish between ordinary annuity and annuity due.	2	2	2

PART – B

S No	Questions	M	BT	CO
1.	Analyze the conflict between profit maximization and shareholder wealth maximization in modern corporations.	10	4	1
2.	Assess the effectiveness of India's regulatory framework (RBI, SEBI, Companies Act) in ensuring financial stability.	10	4	1
3.	Evaluate the role of financial intermediaries in promoting financial inclusion in India.	10	5	1
4.a	Compare the usefulness of balance sheet vs profit and loss statement for investors.	05	2	2
b	Compute Future Value of an amount ₹1,000 invested at 8% interest rate for a period of 5 years.	05	3	2
5.a	Compute Future Value of an amount ₹2,000 invested annually for a period of 6 years at 12% rate of interest.	05	3	2
b	A tech start-up is promised a $1M pay out in 10 years. However, inflation is expected to be 3% for the first 4 years and 5% for the remaining 6 years. Calculate the PV of this pay out in today's dollars.	05	3	2

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Date	16th Apr 2026	Maximum Marks	50+10
Course Code	HS266TEY	Duration	90+20 Min
Sem	VI	CIE – I (Test)	
Universal Human Values – III			

Sl. No.	Quiz	M	BT	CO
1	_____ is central to human existence, _____ is central to _____ existence.	02	1	2
2	Define co-existence.	02	2	1
3	The basic human aspiration is fulfilled through _____ & _____ .	02	1	1
4	_____ from animal consciousness to human consciousness is ensured through _____.	02	2	2
5	Living with human consciousness provides base for _____ & _____ ensuring justice and order leading to _____.	02	1	1

Sl. No.	Test	M	BT	CO
1	Justify the statement "Human Consciousness provides base for Universal Human Order".	10	1	1
2	Explain the concept of basic human aspiration and its fulfillment.	10	2	2
3	Describe Continuous Happiness as the Basic Human Aspiration.	10	3	2
4	Elaborate on personal and societal transformation.	10	3	3
5	Discuss the process of self-exploration in detail.	10	2	2

Course outcomes:

CO 1	Understand the basic human aspiration with program of its fulfillment and meaning of resolution in the complete expanse of human living
CO 2	Understand human being in depth and see how self is central to human being
CO 3	Understand existence in depth and see how coexistence is central to existence
CO 4	Understand human conduct and the holistic way of living leading to human tradition

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	16	34	10	–	16	24	20	–	–	–

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DEPARTMENT OF
HUMANITIES AND SOCIAL SCIENCES

Date : 16.04.2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG- Institutional Elective**	**Duration : 2 hrs**
Course Title: Applied Psychology for Engineers		**Course Code : HS266TEW**

Sl. No	Questions	M	BT	CO
	Part A			
1.1	Who is known as the founder of behaviourism in Applied Psychology?	1	L1	CO1
1.2	Identify the subfield of applied psychology in the following scenario: A company hires a specialist to improve employee motivation and productivity in the workplace.	1	L1	CO1
1.3	Which branch of psychology examines how peer pressure influences individual behaviour in groups?	1	L1	CO2
1.4	In which research method are variables systematically manipulated to observe their effect on behaviour?	1	L2	CO2
1.5	State the main objectives of applied psychology in real-world settings.	2	L2	CO2
1.6	Which field of psychology focuses on assessing and treating emotional and behavioural disorders?	1	L1	CO2
1.7	Rahul is facing anxiety and stress related to his academic performance. Which type of psychologist should he consult?	1	L3	CO4
1.8	Who proposed the concept of primary mental abilities?	1	L1	CO3
1.9	Which ability helps in predicting future performance?	1	L1	CO2
	50 Marks			
	Part B			
2	"Sadie believes that she is the Queen of England and that the CIA is out to get her". Explain in detail the type of psychologist would be most likely to treat her psychological disorder?	10	L1	CO1
3	Explain in detail the approach of psychology that emphasizes conscious experiences, including each person's unique potential for psychological growth and self-direction.	10	L2	CO2
4	Dr. Stephen is going to observe and record children's play behaviour at a nursery school without attempting to influence or control the behaviour. Which method of research is involved? Explain the process and discuss its merits and demerits.	10	L3	CO2
5	Dev shows consistent performance across verbal, numerical, and spatial tasks. How would Spearman interpret this consistency? Justify the statement.	10	L1	CO1
6	A student shows high ability in memory and word fluency but average performance in reasoning tasks. Using Thurston's model, justify this pattern of abilities.	10	L2	CO2

Academic year 2025-2026 (Even Semester)

DEPARTMENT OF
AEROSPACE ENGINEERING

Date	**13th April 2024**	**Maximum Marks**	**60**
Course Code	**AS266TEA**	**Duration**	**120 Min**
Sem	**VI Semester**	**CIE-1**	
FUNDAMENTALS OF AEROSPACE ENGINEERING			

Note: 1. Answer All Questions from both Part-A & Part-B
2. Part-A to be answered in the 1st two pages of the answer booklet

PART – A **(10 Marks)**

	QUESTIONS	M	CO	BL
1.	List the ISA relevance in design and Performance of aircraft	02	CO1	L2
2.	The main source of thrust on large commercial airliners is/are _____.	01	CO1	L2
3.	The pitching motion/moment in an aircraft is controlled by?	01	CO1	L2
4.	What are the three primary control surfaces in an aircraft.	01	CO2	L2
5.	Calculate the geopotential altitude(h) if $r = 6.356810^6$m and h_G = 7 Km	01	CO2	L2
6.	The vertical straight line in the temperature distribution of standard atmosphere graph is called as _____ region	01	CO1	L1
7.	What is the standard sea level value of pressure, temperature and density?	01	CO3	L2
8.	The rudder is located on the _____ of aircraft	01	CO1	L1
9.	Why is the International Standard Atmosphere (ISA) considered a Hypothetical model?	01	CO2	L1

Academic year 2025-2026 (Even Semester)

PART-B

Sl. No.	Questions	M	CO	BT
1	At what value of geometric altitude h_G is the difference h- h_G equal to 2% of h?	10	CO2	L3
2	Identify the major components and its function of a typical Helicopter with a neat sketch.	10	CO3	L2
3	Construct the expression to relate the P, T and ρ along the gradient and isothermal regions of the atmosphere.	10	CO3	L1
4	Calculate the Standard Atmospheric values of P, T and ρ at a geo-potential altitude of 14km.	10	CO1	L2
5	List the various altitudes with a neat sketch and establish a relation expressing the dependence of geometric and geo-potential altitudes.	10	CO4	L1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	16	13	21	10	21	29	10	XX	XX	XX

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DEPARTMENT OF
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Date : 28th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management and Economics		**Course Code : HS261TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Mention any two skills required by an effective manager.	2	L1	01
2	List any two principles of scientific management as proposed by F W Taylor.	2	L1	01
3	Highlight the importance of Hawthorne Studies.	2	L2	01
4	Differentiate between micro and macroeconomics.	2	L2	04
5	What are the two main sectors in a simple circular flow model?	2	L1	05
	Part – B			
1	A retail chain wants to optimize its store operations for better customer service and efficiency. How can the principles of Scientific Management and Administrative Theory help in this scenario?	10	L3	01
2	You are part of the organizing committee for a Techno-Cultural Fest at your institution. Apply the five primary functions of management—Planning, Organizing, Staffing, Directing, and Controlling—to successfully manage the event. Explain how each function can be implemented to ensure the fest runs smoothly and meets its objectives.	10	L3	01
3	A large multinational company is facing challenges in managing its various departments and adapting to changes in the external market environment. Apply **Systems Theory** and **Contingency Theory** to suggest how the company can improve its overall management and operations. Explain how these theories would help in managing the complexity and adapting to the changing conditions.	10	L4	01
4	Consider an economy that is experiencing high inflation and high unemployment simultaneously (stagflation). Using the Circular Flow Model of Economics, explain how inflation and unemployment are interconnected in this economy. How can government policies in the form of fiscal and monetary interventions help address these issues?	10	L4	04
5	A country is transitioning from a command economy to a market economy. Using an overview of Economic Systems, discuss the potential advantages and disadvantages of this shift.	10	L2	05

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Date : 28th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Supply Chain Management		**Course Code : IM362IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List the processes under SRM – macro process.	2	1	CO1
2.	Sketch the implied uncertainty (demand and supply) spectrum and cite examples.	2	2	CO1
3.	Compare Efficient and Responsive Supply Chains(any two).	2	1	CO1
4.	Mention any two challenges to achieving and maintaining strategic fit.	2	1	CO1
5.	Illustrate the relationship between desired response time and number of facilities.	2	2	CO2
	Part – B			
1	Identify the five stages of a supply chain and sketch four process cycles. Explain the sub-processes in a supply chain process cycle of your choice for a manufacturing supply chain.	10	3	CO1
2	Illustrate and analyse the value chain for a manufacturing organisation.	10	3	CO1
3	Evaluate the various stages involved in expanding the scope of strategic fit for an automobile supply chain.	10	3	CO1
4	Discuss the role that information plays in the supply chain, as well as key information-related decisions that supply chain managers must make.	10	2	CO1
5	Select the supply chain of your choice evaluate the performance of a distribution network along two dimensions: Customer needs that are met (service); and Cost of meeting customer needs.	10	4	CO2

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 29th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: ERGONOMICS		**Course Code : IM363IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define human factors engineering in one sentence.	01	L1	CO1
2.	Name any two objectives of ergonomics.	01	L2	CO1
3.	What is meant by user-centred design?	01	L2	CO2
4.	Mention one example of cognitive ergonomics in interface design.	01	L1	CO2
5.	List the three types of anthropometric data.	01	L1	CO2
6.	What does percentile mean in the context of anthropometric measurements?	01	L2	CO2
7.	State one principle of workspace design for sitting tasks.	01	L1	CO3
8.	What is the primary goal of ergonomics in design?	01	L1	CO2
9.	Differentiate between static and dynamic anthropometry with one word each.	01	L2	CO2
10.	Which part of the body is most commonly associated with low back pain due to poor workspace ergonomics?	01	L2	CO3
	Part – B			
1.	Develop a step-by-step ergonomic evaluation plan for a manufacturing workstation. Your answer should include: Identification of key task elements, Application of anthropometric measurements, Postural analysis methods, Recommendations to reduce musculoskeletal risks.	10	L3	CO1
2.	Analyze the causes of work-related musculoskeletal disorders (WMSDs) in the context of poor workspace design. Use the following structure: Ergonomic risk factors, Examples of bad design, Physiological consequences on the musculoskeletal system, Suggestions for redesign based on anthropometry.	10	L4	CO3
3.	A company is designing a standing workbench for both male and female operators in an electronics assembly line. Apply anthropometric principles to determine optimal workbench height, Analyze how human variability influences design decisions, Discuss trade-offs between productivity, comfort, and injury prevention.	10	L4	CO2
4.	The 97.5th percentile value and 2.5th percentile values of normally distributed variable is found by adding/subtracting 1.96 standard deviation from the mean. If the mean body mass of males of a certain populations is 82.1 Kg and the standard deviation is 17.1 kg, what are the 2.5th percentile value and the 97.5th percentile values? How will use the results of this calculation be used in considering the range of users covered in design?	10	L4	CO2

5.	Discuss the scope of work that is covered under the following type of ergonomic studies. i. Physical ergonomics. ii. Cognitive ergonomics. iii. Organizational ergonomics. iv. Macro ergonomics.	10	L2	CO1

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Date : 29th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Human Resource Management & Analytics - IM364TA		

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is Talent Management?	02	L1	CO1
2.	List the type of information that can be collected through job analysis.	02	L2	CO1
3.	List the three steps in Succession planning.	02	L2	CO1
4.	Differentiate between Job specification and Job description.	02	L1	CO1
5.	Name the main internal sources of candidates.	02	L2	CO2
	Part – B			
1	Bright Minds School hired a new HR Manager, Ms. Riya, to transform its HR practices. While she has excellent knowledge of HR policies, she faces challenges in communication, conflict resolution, and strategic thinking. The management is unsure if she is the right fit for a role that requires leadership and organizational development. **Questions:** 1. Based on this case, identify which HR competencies are lacking and how these gaps can affect HRM outcomes. 2. Suggest ways Ms. Riya can develop the necessary competencies to succeed in her role. 3. Discuss how aligning HR objectives with organizational goals can enhance overall institutional effectiveness in this context.	10	L4	CO1
2	Imagine you are working in the HR department of a delivery company called SwiftCart Logistics. The company needs a clear job profile for the role of Delivery Executive to improve hiring and performance. Write the a) Job Description for the role of Delivery Executive and b) Job Specification for the same role.	10	L3	CO1
3	Name any four methods used for collecting Job Analysis information for roles like cashier.	10	L3	CO1
4	Mention two internal and two external sources of recruitment for roles like chefs. Provide one example for each.	10	L3	CO2

5	Ritz Resort is expanding its operations and requires new hires for key positions such as spa therapists, chefs, and front desk receptionists. The HR department is in charge of managing the entire recruitment process to ensure the right candidates are selected quickly and effectively. Questions: 1. Describe the key steps in the recruitment process that the HR team at Ritz Resort should follow when hiring spa therapists and chefs. 2. How can they ensure the quality of candidates selected for these roles?	10	L4	CO2

___ * * * ___

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DEPARTMENT OF
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Date : 02nd May 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Facilities Planning and Design		**Course Code : IM365TDA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Facilities planning are subdivided into the subjects of facilities _____ and facilities _____.	(02)	L1	CO1
2.	State two objectives of facilities planning.	(02)	L2	CO1
3.	List two factors of facilities location.	(02)	L1	CO1
4.	Expand CRAFT and COFAD.	(02)	L1	CO1
5.	What is the basic principle of the CORELAP algorithm, and how is the first department selected?	(02)	L1	CO1
	Part – B			
1.	Enumerate and analyse the characteristics of all the facilities of a supply chain to achieve excellence.	10	L2	CO1
2.	Apply the Winning facilities planning process methodology for facilities planning of a manufacturing organization, and provide your analysis summary.	10	L3	CO3
3.	Perform a comparative assessment of the performance characteristics of process layout, product layout and cellular layout.	10	L2	CO2

<table>
<tr>
<td>4.</td>
<td>An Auto Parts warehouse has requested that a new layout be designed for their main warehouse. The warehouse has 8 major activity centers. The current building area can accommodate 10 X 24 unit squares. Develop the new layout using ALDEP using the data given below.
<table>
<tr><th>Code</th><th>Department Name</th><th>Area (m^2)</th></tr>
<tr><td>1</td><td>Storage</td><td>12800</td></tr>
<tr><td>2</td><td>Counter</td><td>8800</td></tr>
<tr><td>3</td><td>Paint Room</td><td>9200</td></tr>
<tr><td>4</td><td>Receiving</td><td>12000</td></tr>
<tr><td>5</td><td>Office</td><td>8800</td></tr>
<tr><td>6</td><td>Parts Bins</td><td>13200</td></tr>
<tr><td>7</td><td>Tailpipe Racks</td><td>13200</td></tr>
<tr><td>8</td><td>Restroom</td><td>8000</td></tr>
</table>
<table>
<tr><td>1</td><td colspan="7">——-</td></tr>
<tr><td>2</td><td>X</td><td colspan="6">——-</td></tr>
<tr><td>3</td><td>O</td><td>O</td><td colspan="5">——-</td></tr>
<tr><td>4</td><td>A</td><td>X</td><td>U</td><td colspan="4">——-</td></tr>
<tr><td>5</td><td>U</td><td>E</td><td>U</td><td>U</td><td colspan="3">——-</td></tr>
<tr><td>6</td><td>O</td><td>A</td><td>O</td><td>U</td><td>U</td><td colspan="2">——-</td></tr>
<tr><td>7</td><td>O</td><td>I</td><td>I</td><td>U</td><td>U</td><td>I</td><td>——-</td></tr>
<tr><td>8</td><td>O</td><td>O</td><td>O</td><td>U</td><td>O</td><td>O</td><td>O</td></tr>
</table>
<table>
<tr><td>Closeness Value</td><td>A</td><td>E</td><td>I</td><td>O</td><td>U</td><td>X</td></tr>
<tr><td></td><td>64</td><td>16</td><td>4</td><td>1</td><td>0</td><td>-1024</td></tr>
</table>
Minimum Department Preference (MDP) = I = 4, Sweep Width = 2, Number of iterations to be performed, N = 3, Assume Scale: 1 square = 400 m2, Length X Width for the layout: 10 X 24</td>
<td>10</td>
<td>L3</td>
<td>CO3</td>
</tr>
</table>

5.	Given the initial layout, flow matrix and cost matrix shown in figures 1,2 and 3, use the CRAFT scoring method and procedure two ways to obtain an improved layout.(consider any two possible exchanges).	10	L1	CO1

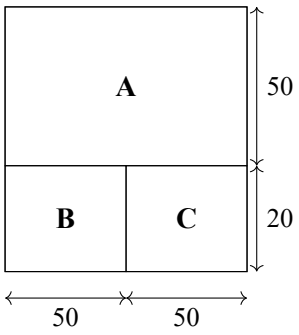


Fig 1. Initial Layout

From \ To	A	B	C
A		4	1
B	2		2
C	2	1	

Fig. 2. Flow Matrix

From \ To	A	B	C
A		1	1
B	1		1
C	1	1	

Fig. 3 Cost Matrix

*****

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 2nd May 2025	**CIE – 1**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the primary goal of Service Operations Management?	2	1	1
2.	Name two major challenges faced by service operations managers in today's dynamic environment.	2	2	2
3.	Differentiate between B2B and B2C services with one example each.	2	3	1
4.	What type of service process is typically used in a fast-food restaurant, and why?	2	3	2
5.	How does service operations management differ from manufacturing operations management?	2	2	2
	Part – B			
1	Identify any 5 tools that have been used on you to collect information to enhance customer satisfaction. Now work backwards and assess what they were trying to understand and how they might have probably improved the service offering. (Use tabular columns and showcase structured assessment)	10	2	2
2	You are a Service Operations Manager of an online retail store. What challenges would you may face as the manager and what measures would you take to overcome them?	10	3	3
3	What are the different types of service processes? Cite the example of any service organization and highlight the different processes involved in that organization.	10	2	2
4	Explain the factors that you will assess as "Service Outcomes" for a service of your choice.	10	2	2
5	Describe a service from both an operational and customer perspective. Assess the mismatches between these perspectives.	5	4	3

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Department of Aerospace Engineering

Date: 02/05/2025	**Test - 1**	**Max. Marks: 50**
Semester: VI	**UG**	**Duration: 90Mins**
Course Title: Fundamental of Aerospace Engineering		**Course Code: AS266TEA**

S No	Questions	M	BT	CO
1.	Mention the effects of atmosphere with respect to design and performance of Air Vehicles	10	L3	CO2
2.	Obtain the relation between Geometric and Geopotential Altitude	10	L2	CO3
3.	Explain briefly the classification of Air vehicles and mention the importance of air vehicles in modern century.	10	L2	CO2
4.	Explain the following concept with application in the aerospace industry. a) Flight Control Surface b) Aircraft Design Configuration c) Modern Aircraft with twin wing configuration	10	L4	CO1
5.	With relevant sketch explain the various parts of Aircraft and Rotorcraft	10	L2	CO1

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DEPARTMENT OF
BIOTECHNOLOGY

Date	**April, 2025**	**Maximum Marks**	**10 + 50**
Course Code	**BT266TEB**	**Duration**	**120 min**
Sem	**VI Semester**	**Closed Book Offline Quiz and Test – 1**	
Healthcare Analytics (Institutional Elective)			

Answer all questions in Part A & Part B, Answer Part A in first two pages in sequence only

Part A

S No	Questions	M	BT	CO
1	There are _____ number of nucleotide bases in DNA	1	2	1
2	Expand DNA	1	2	1
3	Name one primary biological database.	1	1	2
4	What does BLAST stand for?	1	2	2
5	Mention one application of bioinformatics in medicine.	1	1	3
6	Differentiate between DNA and protein sequence databases with examples.	2	3	3
7	How does healthcare data help in improving the lifestyle?	2	4	2
8	Which database provides protein 3D structures?	1	3	2

Part B

S No	Questions (Test)	M	BT	CO
1.	What are the objectives and applications of healthcare data in the current trend?	10	2	3
2	Describe a protein	10	3	3
3	Write short notes on Bioinformatics data.	10	3	3
4	Explain the different types of biological databases used in bioinformatics. How do primary and secondary databases differ? Discuss with examples.	10	3	4
5	What are the advantages and limitations of using these databases for bioinformatics research?	10	2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	28	44	28		28	44	14	14		

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Department of Chemical Engineering

Date: 02.05.2025	**Test - 1**	**Max. Marks: 60**
Semester: 6	**UG**	**Duration: 2 Hrs**
Course Title: Industrial Safety Engineering		**Course Code: CH266TEC**

S No	Questions	M	BT	CO
	PART A			
1	Calculate the risk priority number to identify failure during cash withdrawal in ATM when the severity number is 8, the occurrence number is 5, and the number of times failure is detected during cash withdrawal is 5.	02	2	2
2	Mention four process parameters considered in HAZOP studies.	02	1	1
3	The guide word used in HAZOP for qualitative modification/increase is _____.	01	1	1
4	What do the abbreviations UCIL and MIC stand for?	02	1	2
5	In which years did the Bhopal Gas Tragedy and the Chernobyl Disaster occur?	02	1	2
6	The Bhopal Gas Tragedy occurred due to a _____ reaction.	01	1	2
	PART B			
1	Explain the failure mode and effect analysis procedure in detail with an example.	10	2	3
2a	Using five relevant guide words, perform HAZOP analysis for a shell and tube heat exchanger in which hot liquid flows at the shell side and cold liquid on the tube side.	05	3	2
2b	Write the flow diagram for the sequence of the HAZOP study.	05	2	2
3a	Explain the risk matrix in detail with a matrix table.	05	2	2
3b	Briefly explain the safety and health issues, supported by relevant examples and an appropriate graphical representation.	05	2	2
4	Explain the Bhopal Gas Tragedy with reference to: 1. The incident overview 2. Onsite safety measures with a relevant sketch to illustrate the scenario.	10	3	3
5	Provide a detailed explanation of the Jim Walter Resources Mine Disaster, including a timeline of the key events and a diagrammatic sketch illustrating the sequence of the mishap.	10	3	3

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Department of Computer Science and Engineering

Date: May 2025	Test - 1	Max. Marks: 60
Semester: VI	UG	Duration: 2 Hrs
Course Title: ROBOTIC PROCESS AUTOMATION		Course Code: CS266TED

PART A

S No	Questions	M	BT	CO
1.	What is the primary difference between traditional automation and RPA?	01	1	1
2.	Mention types of tasks that are ideal for RPA.	02	2	1
3.	Name any two industries where RPA is commonly used.	01	2	1
4.	RPA bots interact with applications through the _____.	01	1	1
5.	A _____ is used in RPA to repeat a sequence of actions.	01	1	2
6.	List the type of variables in UiPath.	02	2	1
7.	Mention two differences between sequences and flowchart.	02	2	2

PART B

S No	Questions	M	BT	CO
1. a	Discuss different types of RPA robots mention various features.	05	2	2
b	Provide benefits of using RPA also discuss the business implementations.	05	3	2
2. a	What are generic value variables in RPA? Provide examples.	05	2	3
b	Differentiate between variables and arguments with an example	05	3	3
3. a	Explain Center of Excellence (CoE) with respect to RPA and its types	10	2	2
4. a	Describe the RPA development life cycle	06	2	1
b	List the effective approaches in selecting processes for automation.	04	2	1
5. a	Differentiate between waterfall and agile methodologies	04	2	2
b	What are namespaces? Discuss with a use case.	06	2	2

Course Outcomes

CO1	Understand RPA principles, its features and applications
CO2	Demonstrate proficiency in handling variables and decision making inside a workflow and data manipulation techniques.
CO3	Gain insights into recording, email automation and exception handling and orchestrator
CO4	Analyze the trends in automation and choose business strategy to design a real world automation workflow.

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Department of Civil Engineering

Date: 02-05-2025	**CIE - 1**	**Max. Marks: 60**
Semester: VI	**UG**	**Duration: 2 Hrs**
Course Title: Intelligent Transport Systems		**Course Code: CV266TEE**

PART A

S No	Questions	M	BT	CO
1.	Mention two key characteristics of a transport system.	2	2	1
2.	State any two major transport problems in urban areas.	2	2	1
3.	How does urbanisation influenced transportation needs?	2	2	2
4.	Mention two challenges in implementing ITS in developing countries.	2	2	1
5.	State any two reasons why ITS is important for the Indian transport system.	2	2	2

PART B

S No	Questions	M	BT	CO
1.	Outline ten key critical areas where training and educational needs for Intelligent Transport Systems (ITS) are identified according to the National ITS Professional Capacity Building Program.	10	3	1
2.	Identify and describe the issues related to the transportation system within a city.	10	3	1
3.	Describe the concept of Intelligent Transport Systems (ITS) and outline its user services.	10	3	2
4.	How does ITS fit within the traditional transportation system? What are the difference and synergies between traditional transportation systems and ITS?	10	3	1
5.	Identify the most congested corridor in your area. What are the problems associated with that corridor? What type of technology can help solve those problems?	10	4	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	36	24	**	**	**	10	40	10	**	**

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Department of Chemistry

Date: 02-05-2025	**Test - 1**	**Max. Marks: 50**
Semester: 6	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
Course Title: ADVANCED ENERGY STORAGE DEVICES FOR E-MOBILITY		**Course Code: CM266TEG**

S No	Questions	M	BT	CO
1	Differentiate between Battery Electric Vehicles (BEVs), Hybrid Electric Vehicles (HEVs), and Plug-in Hybrid Electric Vehicles (PHEVs) with respect to their energy requirements and functionalities along with their advantages and limitations.	10	3	4
2	Describe the construction and working of a Lithium Cobalt Oxide battery. Discuss its advantages and limitations in the context of e- mobility.	10	4	1
3	Enumerate the desirable battery characteristics for use in electric vehicles. Explain the various battery characteristics of advanced batteries how these parameters will help in designing advanced battery for e mobility.	10	4	3
4	a) Write short notes on the characteristics of anode materials in determining lithium-ion battery performance. b) A battery is designed using Zn as anode and Cu as cathode. Calculate the EMF of a cell at 30 °C, when concentration of products and reactants are 5M and 0.5M respectively. (Given:(F=96500C) Reduction potentials of Zn and Cu are -0.76 V and 0.34 V respectively). What may be the capacity of this cell if its weight is 0.6 kg and molar mass is 30?	10	3	2
5	a) How batteries are classified? Explain each classification with an example each. b) How battery electrolytes are classified explain with example each. Explain the characteristics of electrolytes in advanced lithium battery.	10	1	1

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Department of Electronics and Communication Engineering

Date: 2.5.2025	**Quiz - I**	**Max. Marks: 10**
Semester: VI	**UG**	**Duration: $\frac{1}{2}$ Hrs**
Course Title: Human Machine Interface		**Course Code: EC266TEH**

S No	Quiz -I	M	BT	CO
1.	What is the role of forward collision warning in driver safety?	1	1	1
2.	What is the purpose of Traffic Sign Recognition (TSR) in vehicles?	1	1	1
3.	List any two ergonomic factors to consider in HMI design.	1	1	1
4.	_____ Software is commonly used for designing automotive HMIs?	1	1	1
5.	Adaptive Cruise Control HMI helps the driver with_____	1	1	1
6.	Why CAN bus use a differential signalling?	1	2	1
7.	What is the significance of terminating resistors in CAN?	1	2	3
8.	Name two common display/HMI elements shown when ACC is active.	1	1	2
9.	_____ is an example of vehicle-to-Smartphone connectivity.	1	1	1
10.	_____ technology enables hand gesture detection in automotive HMIs.	1	1	1

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Department of Electronics and Communication Engineering

Date: 2.5.2025	**Test - 1**	**Max. Marks: 50**
Semester: VI	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
Course Title: Human Machine Interface		**Course Code: EC266TEH**

S No	Questions	M	BT	CO
1.a.	Discuss any four advantageous of Human Machine Interface.	4	1	1
1.b.	With a neat block diagram, describe Human machine Interface Architecture.	6	1	1
2.a.	Discuss the different stages of HMI interface problem solving based on Human Cognitive Psychology.	4	2	1
2.b.	With a neat block diagram, describe HMI Framework model with CGI Architecture.	6	3	2
3.a.	Elaborate the different layers of In vehicle Infotainment system with a neat diagram.	5	2	2
3.b.	Illustrate the principle of parking-gap measurement.	5	2	2
4.a.	Elaborate the different components of Automotive infotainment systems.	4	1	1
4.b.	How does the nodes interconnected via a CAN bus to achieve bus arbitration?	6	3	2
5.a.	Explain all the message formats in CAN.	5	2	1
5.b.	With a neat block diagram describe the basic structure of Adaptive Cruise Control.	5	1	1

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 29th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Elements of Financial Management		**Course Code : IM266TEQ**

Note:
1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.
3. Use of Interest Factor table permitted.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Mention any one distinction between Financial Accounting and Financial Management in terms of its scope.	01	L_1	CO1
2.	What is the main function of the Reserve Bank of India?	01	L_2	CO1
3.	What is the full form of NABARD?	01	L_1	CO1
4.	The National Stock Exchange (NSE) is located in which city?	01	L_1	CO1
5.	Which body regulates the insurance sector in India?	01	L_2	CO1
6.	What is the meaning of the term Financial intermediaries?	01	L_1	CO2
7.	What is a Fixed asset?	01	L_2	CO2
8.	Give any one example for Non-current assets.	01	L_2	CO2
9.	Define Time Value of Money.	01	L_1	CO2
10.	State the formula for the present value of an annuity.	01	L_2	CO2
	Part – B			
1.	Discuss the functions performed by financial markets. What are the different ways of classifying financial markets? Write briefly on primary market.	10	L_3	CO1
2.	What are the alternative key objectives or goals of Financial Management? Why shareholder's wealth maximization /value maximization is considered as better objective of financial management instead of profit maximization?	10	L_4	CO1
3 a.	Emerson Cammack wishes to purchase an annuity contract that will pay him ₹7,000 a year for the rest of his life. The Philo Life Insurance Company figures that his life expectancy is 20 years, based on its actuary tables. The company imputes a compound annual interest rate of 6 percent in its annuity contracts. How much will Cammack have to pay for the annuity? .How much would he have to pay if the interest rate were 8 percent.	05	L_3	CO2
b.	₹100 is received at the end of one year, ₹500 at the end of two years, and ₹1,000 at the end of three years. What is the aggregate present value of these receipts, assuming a discount rate of (i) 4 percent? (ii) 25 percent?	05	L_3	CO2

4.	Discuss the three broad areas of financial decision making. Give one example for each of these areas by taking the case of any organization in your respective domain of technology. Discuss by briefly describing the work that the organization does and relate it to the three areas of decision making.	10	L_4	CO1
5 a.	If you deposit, ₹1000 today in a bank which pays 10 percent interest compounded annually how much will the deposit grow to after 8 years and 12 years.	05	L3	CO2
b.	Mr. Vinay plans to send his son for higher studies abroad after 10 years. He expects the cost of these studies to be ₹1,000,000 at the end of 10 years. If the interest rate is 12 percent.	05	L3	CO2

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Department of Mathematics

Date: 02/05/2025	**CIE - 1**	**Max. Marks: 100**
Semester: VI	**UG**	**Duration: 2 Hrs. (2pm-4pm)**
Course Title: Mathematical Modelling		**Course Code: MA266TEU**

S No	Quiz: Questions	M	BT	CO
1	In a certain country, the growth in the number of computer scientists is observed to follow the law $\frac{dx}{dt} = \frac{1}{15x}$. If initially there were 1000 of them, then after 15 years, their approximate number will be _____ and their doubling period is approximately _____ years.	2	L1	2
2	A body of mass 3 kg is projected at an angle of 30° from the ground with an initial velocity of 25 m/s, acceleration due to gravity is $g \simeq 10\ m/s^2$. The maximum horizontal distance is _____.	2	L1	1
3	A cup of coffee at 90°C is left in a room at 25°C. If the cooling constant is $k = 0.05$, using Newton's law of cooling, the temperature of the coffee after 10 minutes is _____.	2	L2	3
4	Find the maximum height of the projectile at the time $v\,\frac{\sin\alpha}{2g}$.	2	L2	2
5	Discuss about Fick's law of diffusion model.	2	L1	1

Test

SL. No	Test	M	BT	CO
1.	Develop a differential equation model based on the logistic law of population growth, solve it and analyse any one real life scenario.	10	L3	2
2.	Describe the systematic 12-step procedure for tackling a problem through the use of mathematical modelling.	10	L1	1
3.	Model and describe the second-order differential equations representing a mass-spring-dashpot mechanical system. Also highlight the analogies between the elements of the mechanical and electrical systems.	10	L2	2
4.	A nutrient is periodically injected into a patient's bloodstream at fixed intervals of time T. Meanwhile, the nutrient is continuously absorbed by the body at a rate proportional to its concentration $c(t)$ in the blood at time t. Derive the differential equation governing $c(t)$. Solve it and interpret the long-term behaviour of the nutrient concentration in the bloodstream.	10	L3	2
5.	i) Model the motion of a projectile launched in a vacuum with an initial velocity at a given horizontal angle. Formulate the governing equations of motion, solve them and determine key characteristics such as time of flight, maximum height and horizontal range. ii) Explain any two characteristics of Mathematical modelling.	10	L3	2

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Department of Mathematics

Date: 02-05-2025	**CIE 1**	**Max. Marks: 10+50**
Semester: VI	**UG**	**Duration: 2 Hrs (2PM- 4PM)**
Course Title: Mathematics of Quantum Computing		**Course Code: MA266TEV**

S No	Quiz	M	BT	CO
1.	What is the trigonometric and exponential form of $z = -3 + 3i$?	2	1	1
2.	Given the vector $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and matrix $P = \begin{bmatrix} 2 & -1 \\ -1+2i & 3 \\ -3 & -3+i \end{bmatrix}$, the matrix P transforms the vector v into _____.	2	1	1
3.	Find the norm of the vector $v = \begin{bmatrix} i \\ 2+i \\ -3 \end{bmatrix}$.	2	2	2
4.	Show that the Unitary transformations preserve inner products.	2	3	3
5.	If the Pauli-X gate is applied to the qubit state $\|\psi\rangle = \alpha\|0\rangle + \beta\|1\rangle$, what is the resulting state?	1	2	3
6.	A quantum state $\|\psi\rangle = \frac{1}{\sqrt{3}}\|0\rangle + \sqrt{\frac{2}{3}}\beta\|1\rangle$ is measured in the computational basis $\{\|0\rangle, \|1\rangle\}$. What is the probability of obtaining $\|1\rangle$?	1	2	2
S No	**Test**	**M**	**BT**	**CO**
1a	Decompose the vector $\vec{v} = \begin{bmatrix} 3+i \\ 1-i \end{bmatrix}$ into the orthonormal basis $\left\{ \vec{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix}, \vec{u}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} \right\}$.	5	2	1
1b	Given the quantum states: $\|v\rangle = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ i \\ -1 \end{bmatrix}$, $\|w\rangle = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -i \\ 2 \end{bmatrix}$, show that $\langle w \,\|\, v\rangle = \overline{\langle v \,\|\, w\rangle}$.	5	1	2
2	Given the quantum state: $\|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix}$, calculate the probability of measuring it in the $\|+\rangle$ state using (a). Born's rule and (b). basis expansion.	10	3	2
3	State the no-cloning theorem and explain its implications for quantum computing. Consider a qubit in an unknown state $\|\psi\rangle = \alpha\|0\rangle + \beta\|1\rangle$. Demonstrate why a unitary operation cannot clone this state by attempting to construct a cloning operation and showing the contradiction.	10	3	2
4	Determine whether the following states are separable: (i) $\|\Phi\rangle = \frac{1}{2}(\|00\rangle + \|01\rangle - \|10\rangle - \|11\rangle)$ (ii) $\|\Psi\rangle = \frac{1}{\sqrt{2}}(\|01\rangle + \|10\rangle)$	10	3	3

5	Suppose a square matrix U is given by $U = \begin{pmatrix} v_1^1 & v_1^2 & \cdots & v_1^n \\ v_2^1 & v_2^2 & \cdots & v_2^n \\ \vdots & \vdots & \ddots & \vdots \\ v_n^1 & v_n^2 & \cdots & v_n^n \end{pmatrix}$. Let $v^i = (v_1^i \; v_2^i \; \cdots \; v_n^i)^T$ represent i-th n-dimensional column vector and $v_i = (v_i^1 \; v_i^2 \; \cdots \; v_i^n)$ represent i-th n-dimensional row vector of matrix U. This means that $U = (v^1 \; v^2 \; \cdots \; v^n) = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$. Show that the following are equivalent statements about a matrix U: (i) $U \cdot U^\dagger = I_{n\times n}$ (ii) $\{v_1, v_2, \ldots, v_n\}$ is an orthonormal set.	10	3	3

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DEPARTMENT OF
Humanities and Social Sciences

Date : 02/05/2025	**CIE– I**	**Max. Marks : 10 + 50**
Semester : VI	**UG-Institutional Elective**	**Duration : 30 + 90 Min**
Course Title: Applied Psychology For Engineers		**Course Code : HS266TEW**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What are the goals of Applied Psychology?	2	L1	CO1
2.	How does the training of psychologists differ from that of psychiatrists?	2	L1	CO2
3.	What do you mean by slip of the tongue and free association?	2	L2	CO2
4.	Who proposed that the g factor represents the highest-order common factor among individual differences in IQ?	2	L3	CO3
5.	Sandra is 8 years old and, according to the Stanford Binet, he has a mental age of 10. What is his IQ?	2	L4	CO4
	Part – B			
1 a.	Explain the differences in the role of Clinical and Industrial Psychologists from today's perspective.	6	L2	CO2
b.	Describe the nature of Applied Psychology	4	L2	CO1
2a.	Differentiate between the Questionnaire and Inventory method.	2	L1	CO1
b.	Prepare a questionnaire on students' academic stress in the Autonomous scheme in an engineering institution consisting of 10 items on 3- point rating scale with interpretation.	8	L3	CO2
3 a.	Is child psychology different from developmental psychology? Describe.	4	L1	CO1
b.	Dr.Mark explains to a client that his feelings of hostility toward a co-worker are most likely caused by the way the client interprets the coworker's actions and the way he feels that people should behave at work. What perspective is Dr. Mark most likely working from? Explain in detail.	6	L1	CO3
4	What do you understand by the term Intelligence? Explain Thurston and Spearman theory of Intelligence merits and de-merits by illustrating an example/ experiment	10	L3	CO3
5 a.	Explain the differences between Intelligence, Aptitude and Achievement.	3	L1	CO1
b.	If fluid and crystallized intelligence do exist, how might they be tested? In view of this explain Guilford theory of intelligence.	7	L2	CO2

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Department of Civil Engineering

Date	**2nd May 2025**	**Maximum Marks**	**50+10**
Course Code	**HS266TEY**	**Duration**	**90+20 Min**
Sem	**VI**	**CIE – I (Test)**	
Universal Human Values – III			

S No	**Quiz**	**M**	**BT**	**CO**
1	The basic human aspiration is _____ & _____ for humans alone may suffice for animals but _____ & _____ are the highest order in dynamic activity of _____ self.	02	1	1
2	Living with human consciousness provides the base for _____ & _____.	02	2	1
3	Humane society is ensured with _____ & _____.	02	1	2

S No	**Test**	**M**	**BT**	**CO**
1	Elucidate on basic human aspiration and its fulfillment.	10	1	1
2	Explain the role of education-sanskar to facilitate transition from Animal to Human Consciousness.	10	2	2
3	Highlight the expected transformations when living with human consciousness.	10	3	2
4	Elaborate on societal transformation (inhuman → human).	10	3	3
5	Discuss the concept of self-exploration as the methodology.	10	2	2

Course outcomes:

CO 1	Understand the basic human aspiration with program of its fulfillment and meaning of resolution in the complete expanse of human living
CO 2	Understand human being in depth and see how self is central to human being
CO 3	Understand existence in depth and see how coexistence is central to existence
CO 4	Understand human conduct and the holistic way of living leading to human tradition

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	16	34	10	–	16	24	20	–	–	–

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